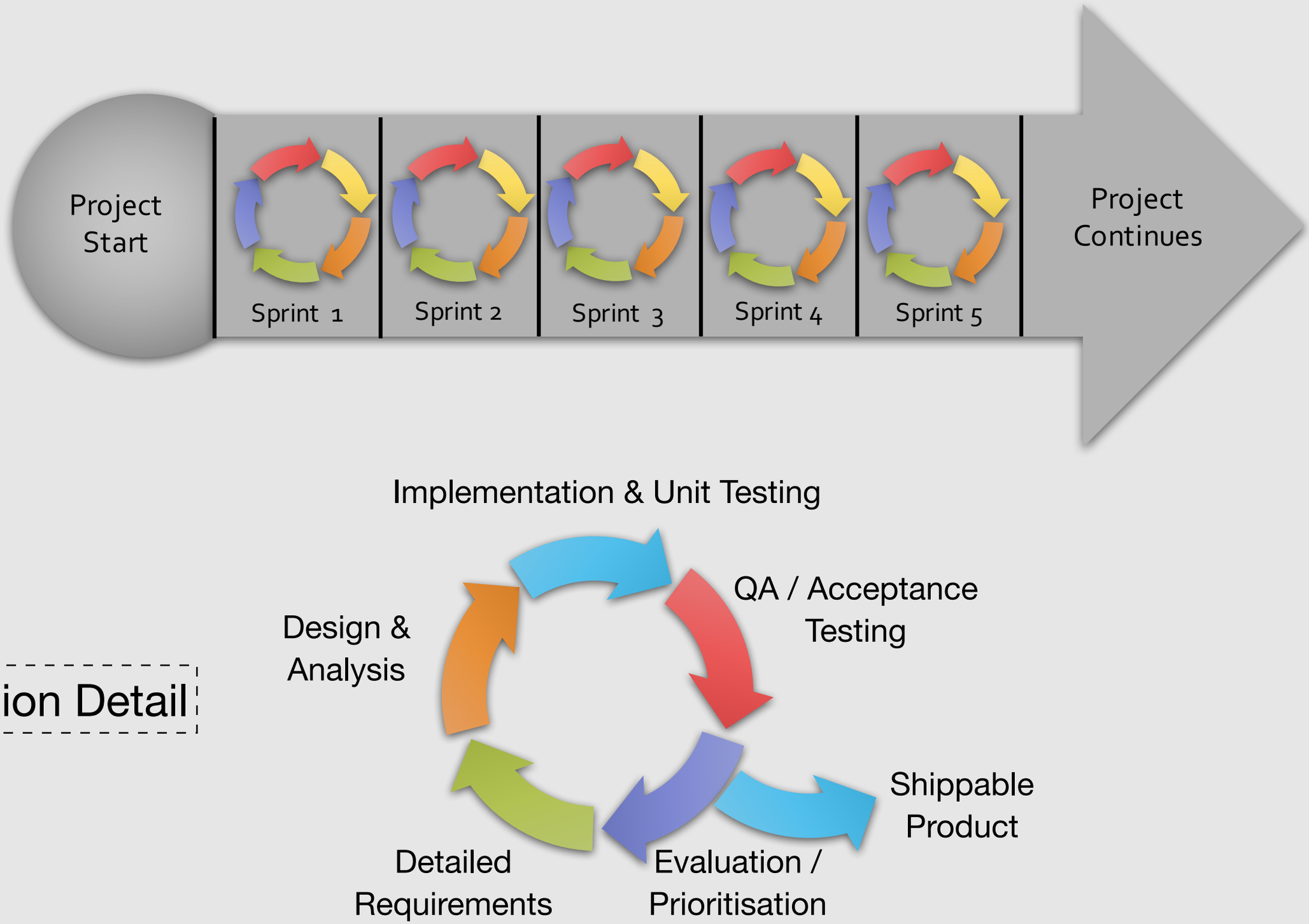
Agile Scrum

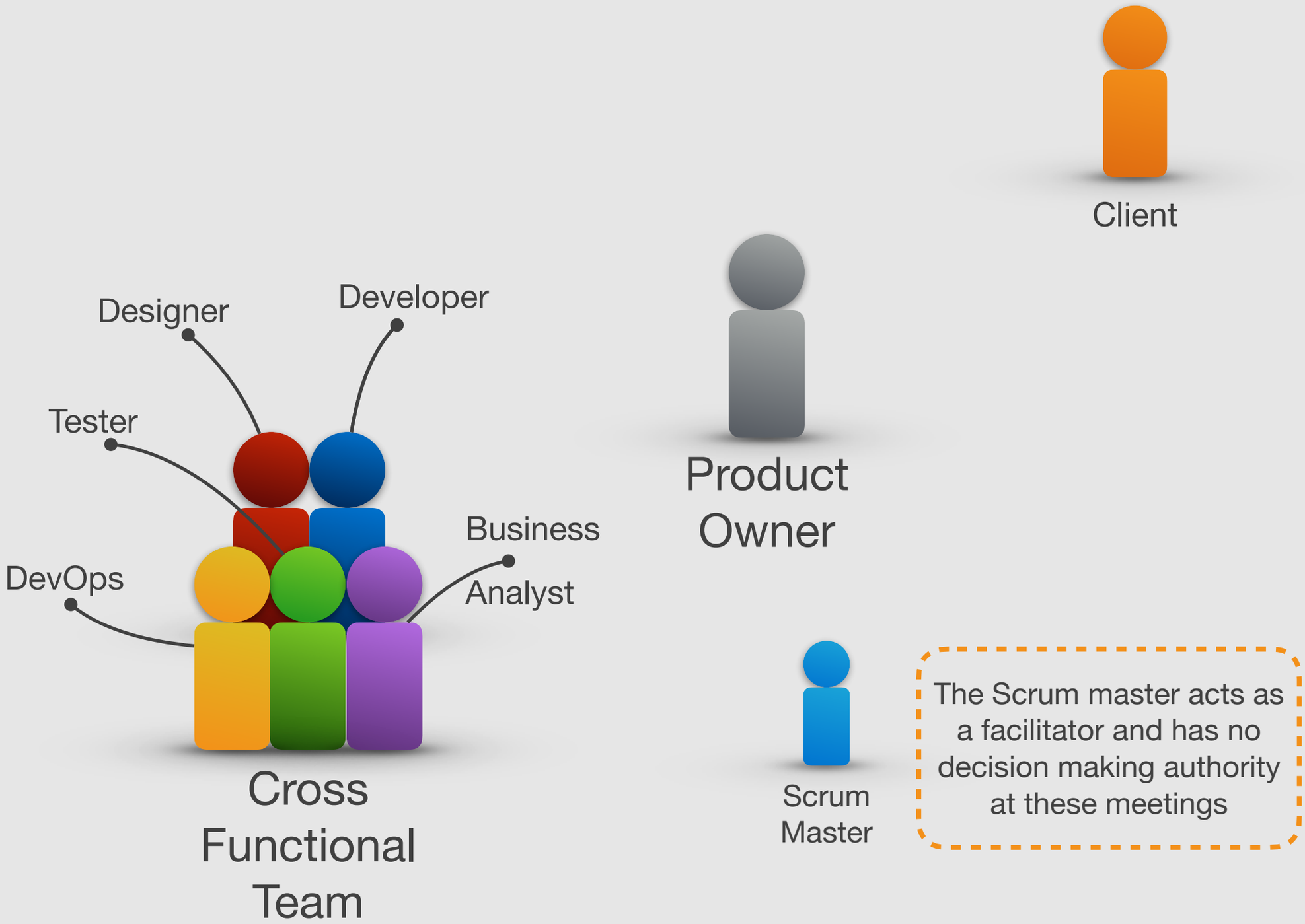
Software Product and Process Management

Introduction to Scrum

Scrum Methodology



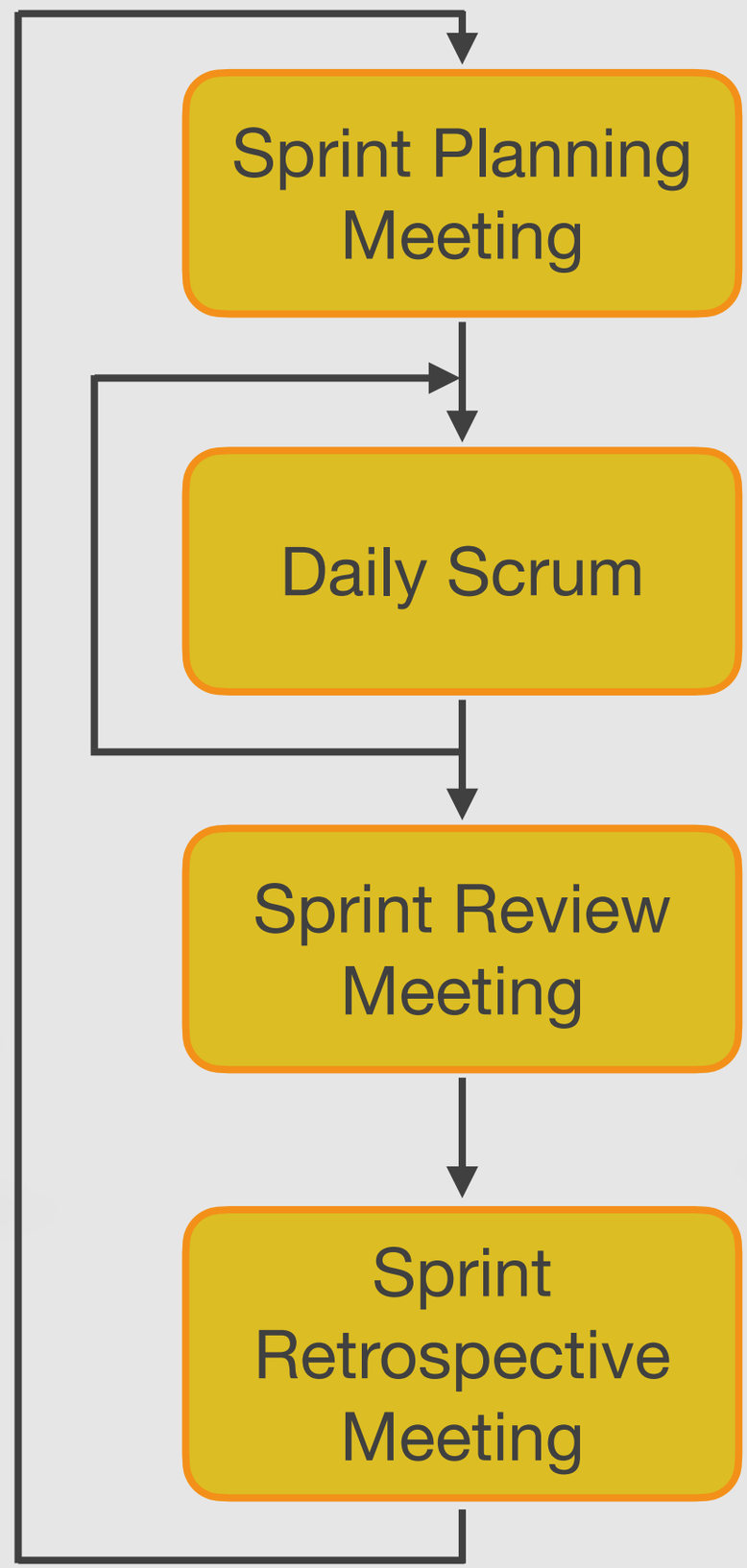
Agile Scrum - People



Agile Scrum - Ceremonies

Product Owner

Cross Functional Team

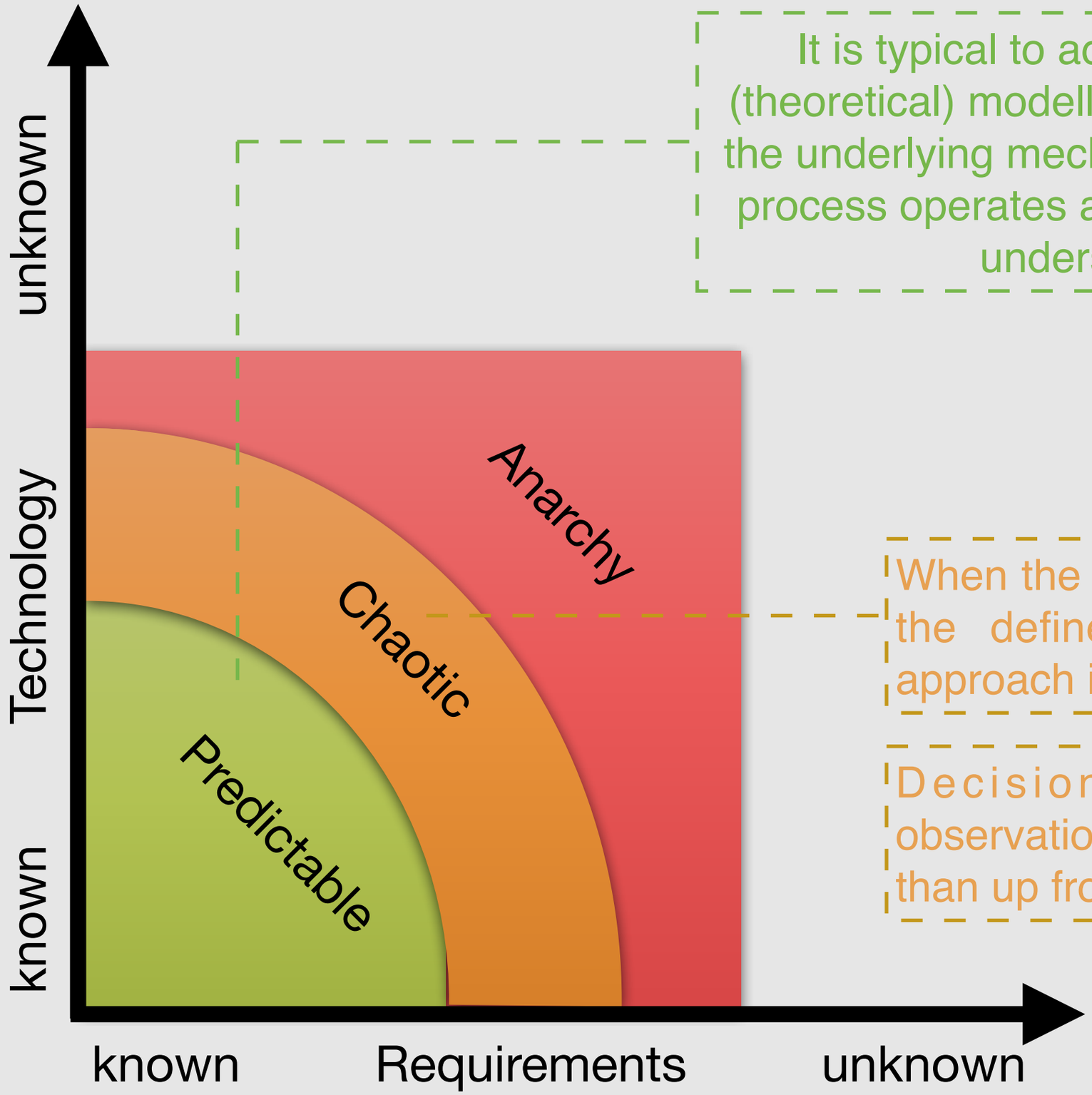


Backlog Refinement Meeting

Scrum Master

The Scrum master acts as a facilitator and has no decision making authority at these meetings

When to use Scrum

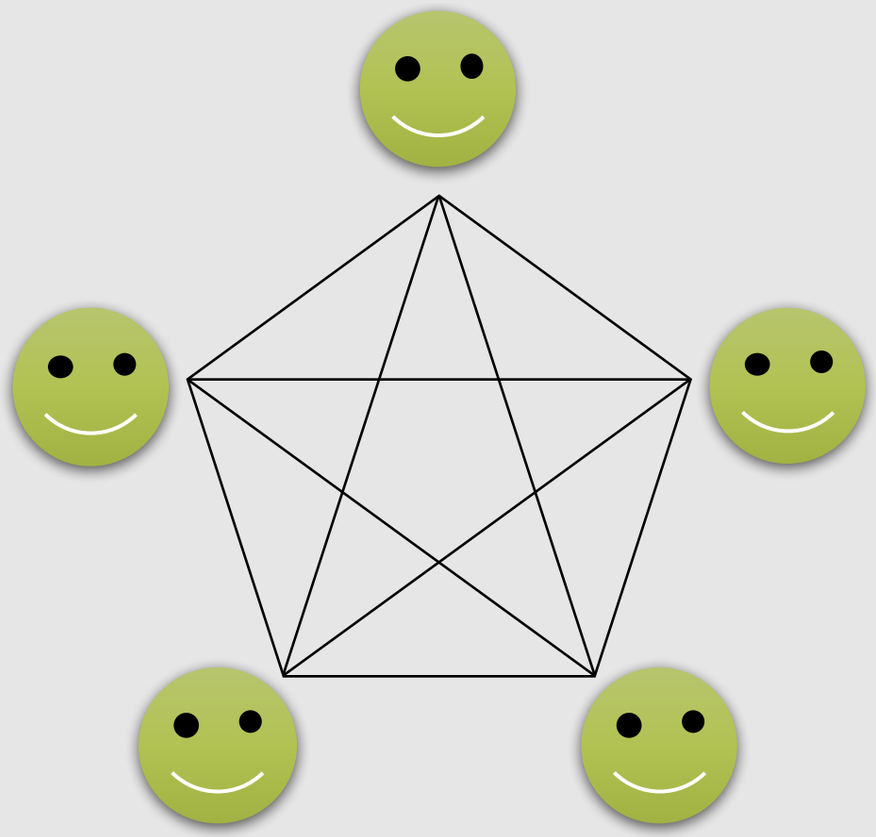


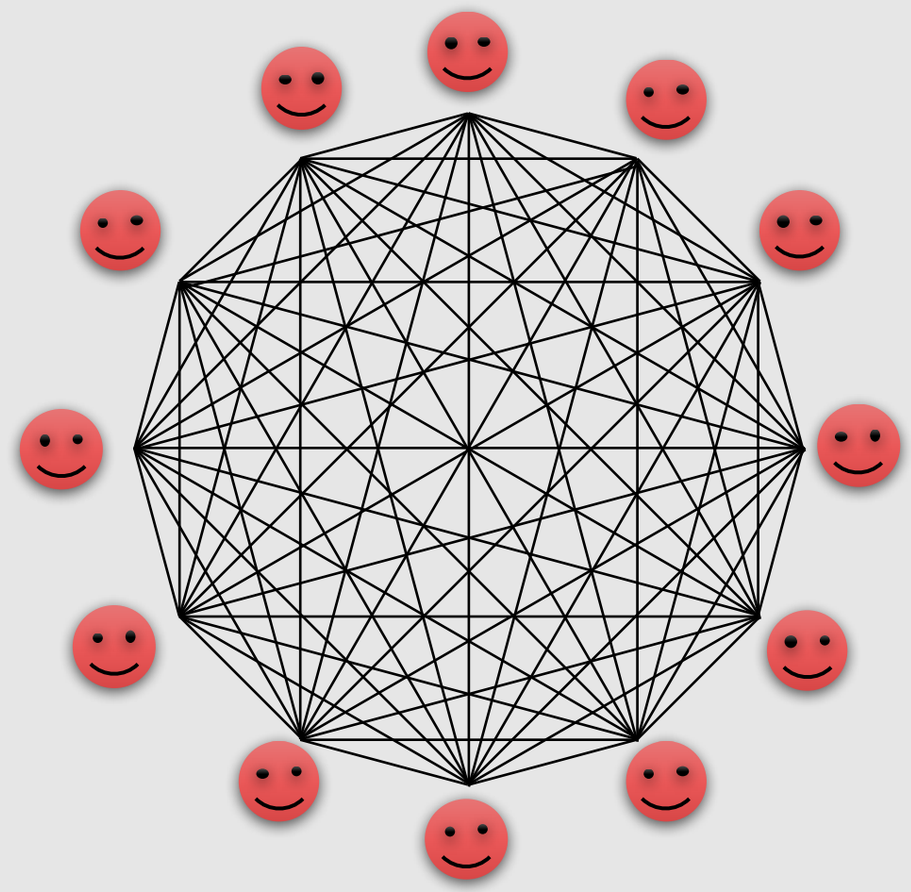
It is typical to adopt the defined (theoretical) modelling approach when the underlying mechanisms by which a process operates are reasonably well understood.

When the process is too complicated for the defined approach, the empirical approach is the appropriate choice.

Decisions are made based on observation and experimentation rather than up front planning.

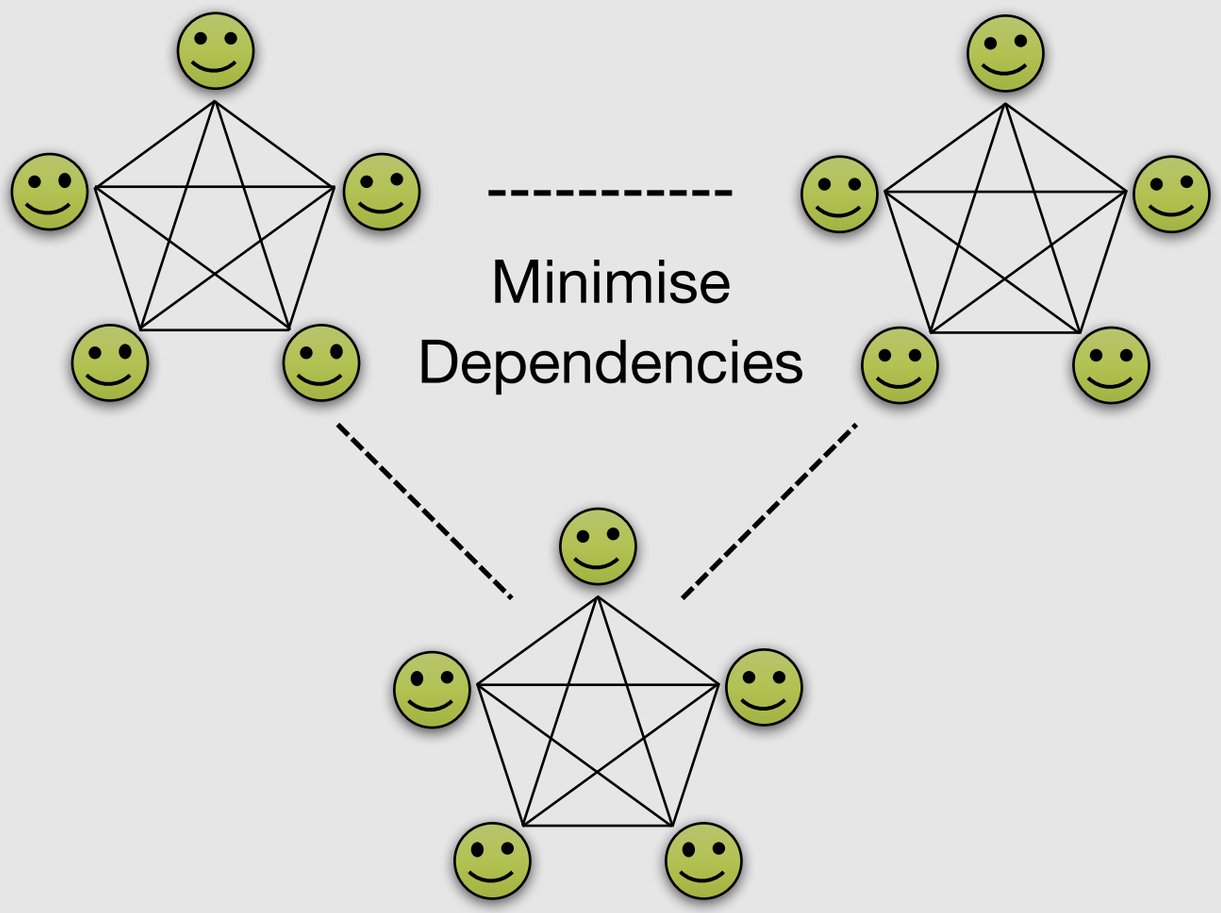
Communication & Scaling





$$\text{number of channels} = \frac{n\,(n-1)}{2} = \frac{n^2 - n}{2}$$

Scaling at Large Organisations



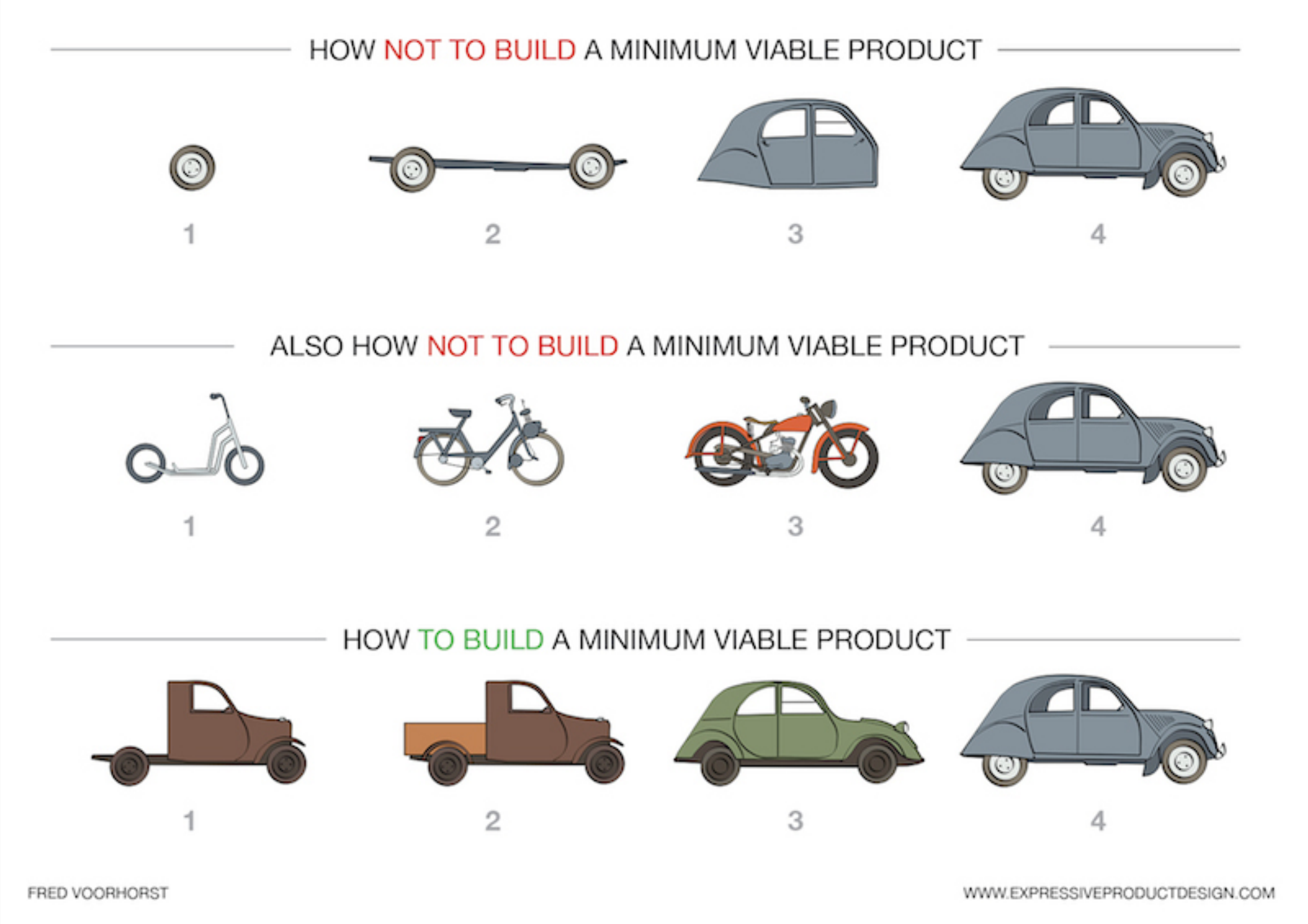
Spotify Engineering Culture

- Part I – 13 minutes
  - https://vimeo.com/85490944
- Part II – 13 minutes
  - https://vimeo.com/94950270

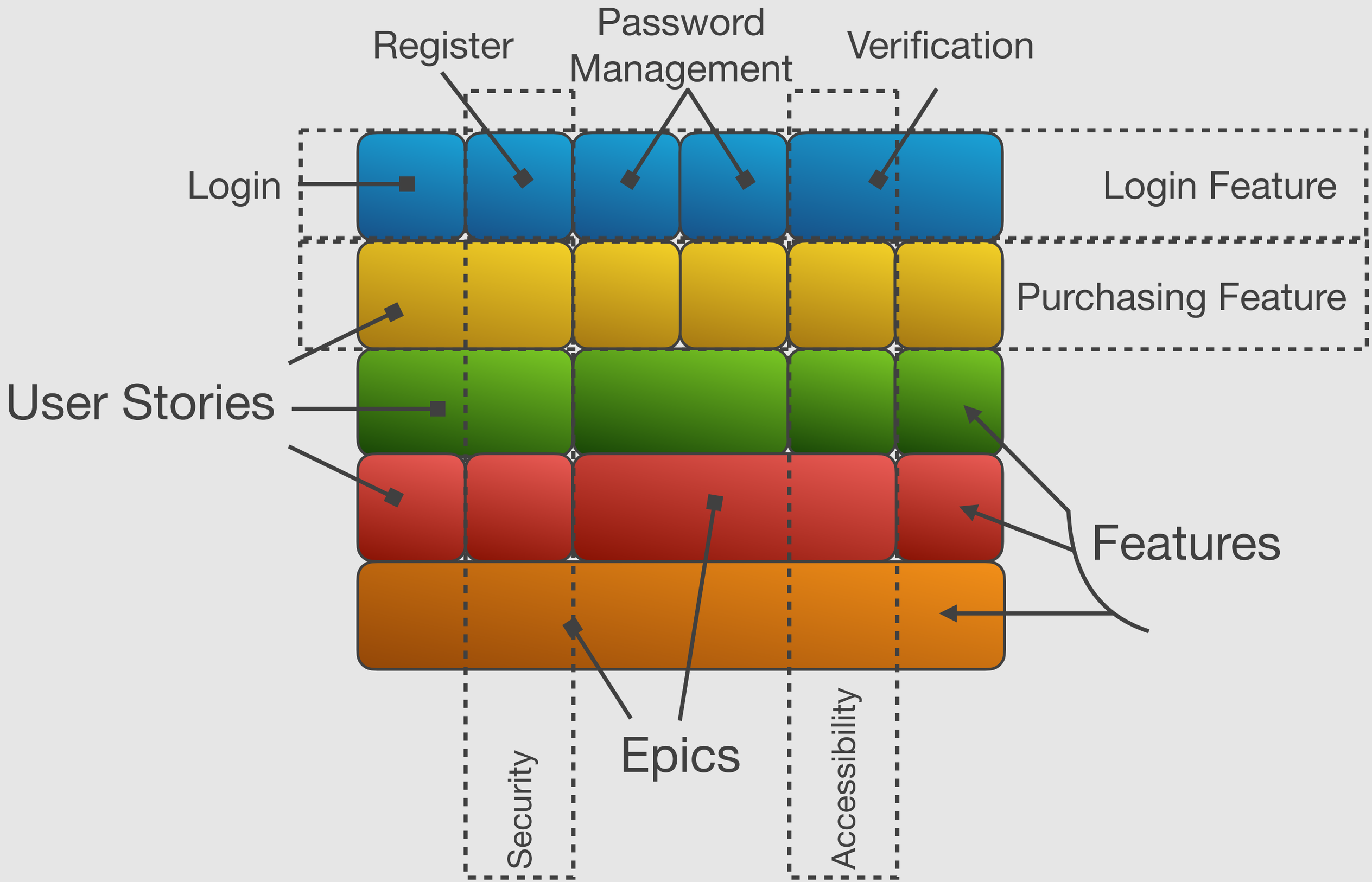
Example of an Agile Environment

A pod of workers at Electronic Arts in Montreal. The studio is now working on the sequel to Army of Two. (eamontreal.com)

How to Build a MVP



Themes, User Stories & Epics



Kanban

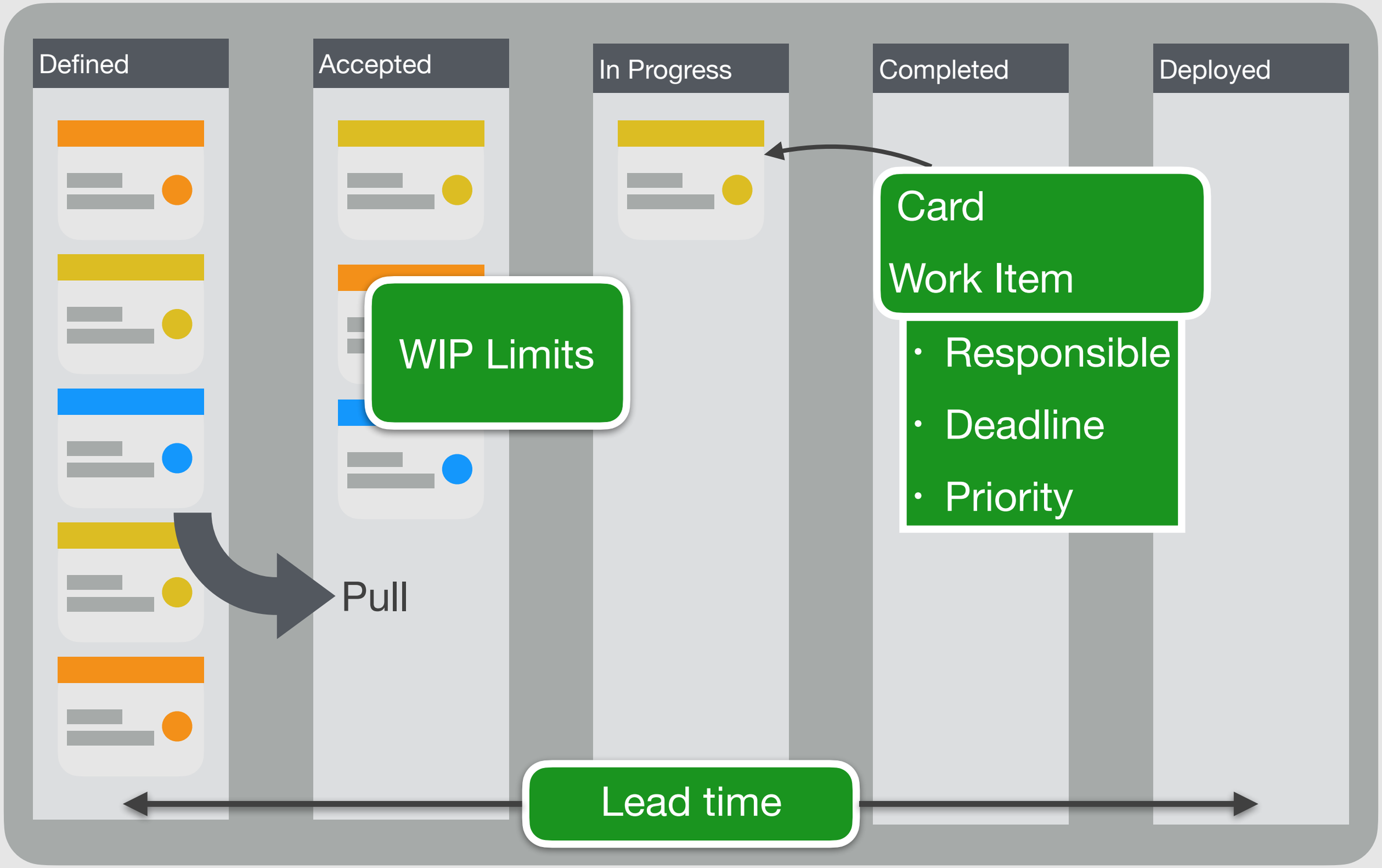
Visual Board

Kanban Practices

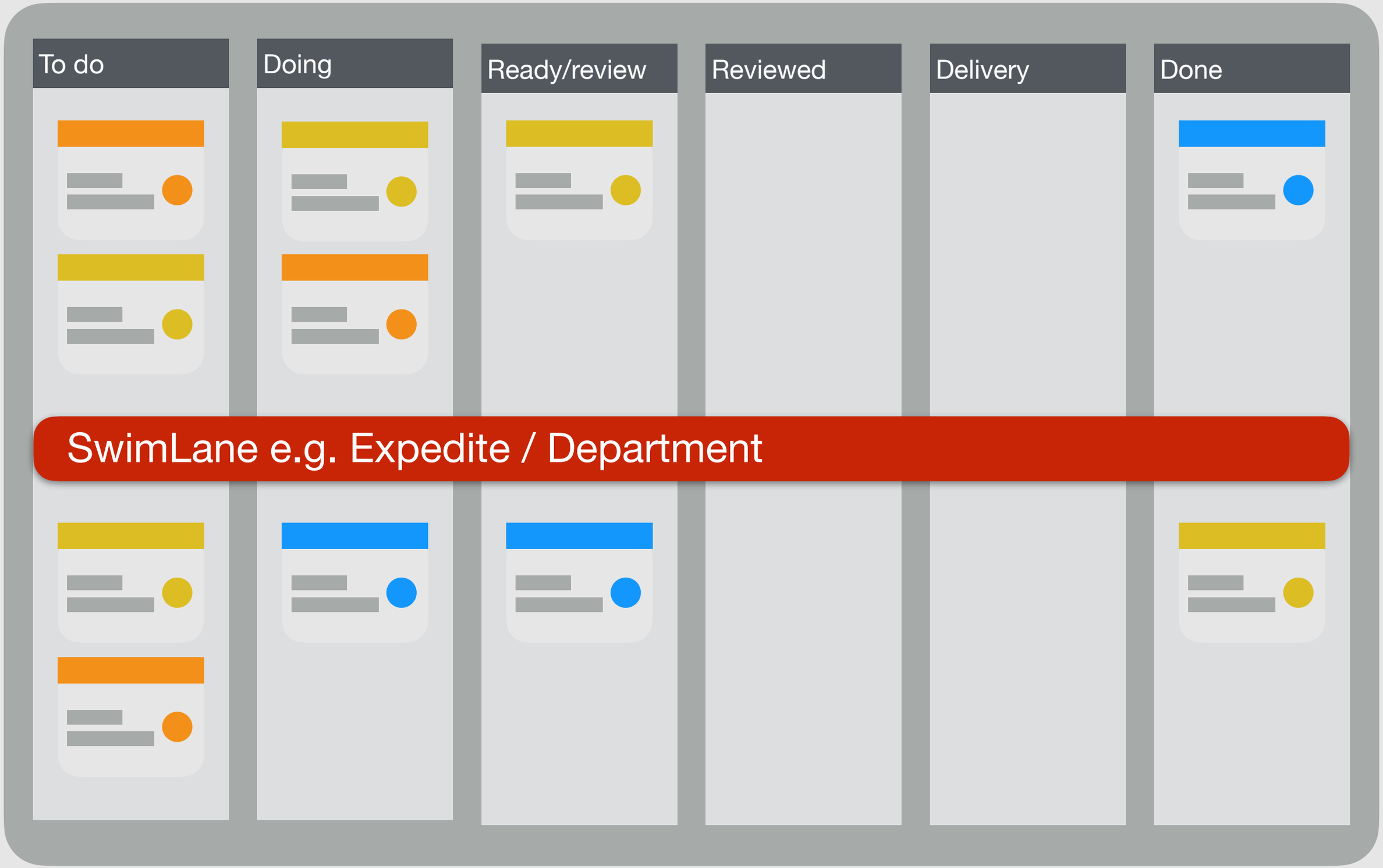
1. **Visualise the work flow**
2. **Limit the WIP**
3. **Manage Flow**
4. Explicit policies
5. Implement feedback loops
6. Improve collaboratively

https://teamhood.com/kanban-resources/

Kanban Board - Visualise work flow

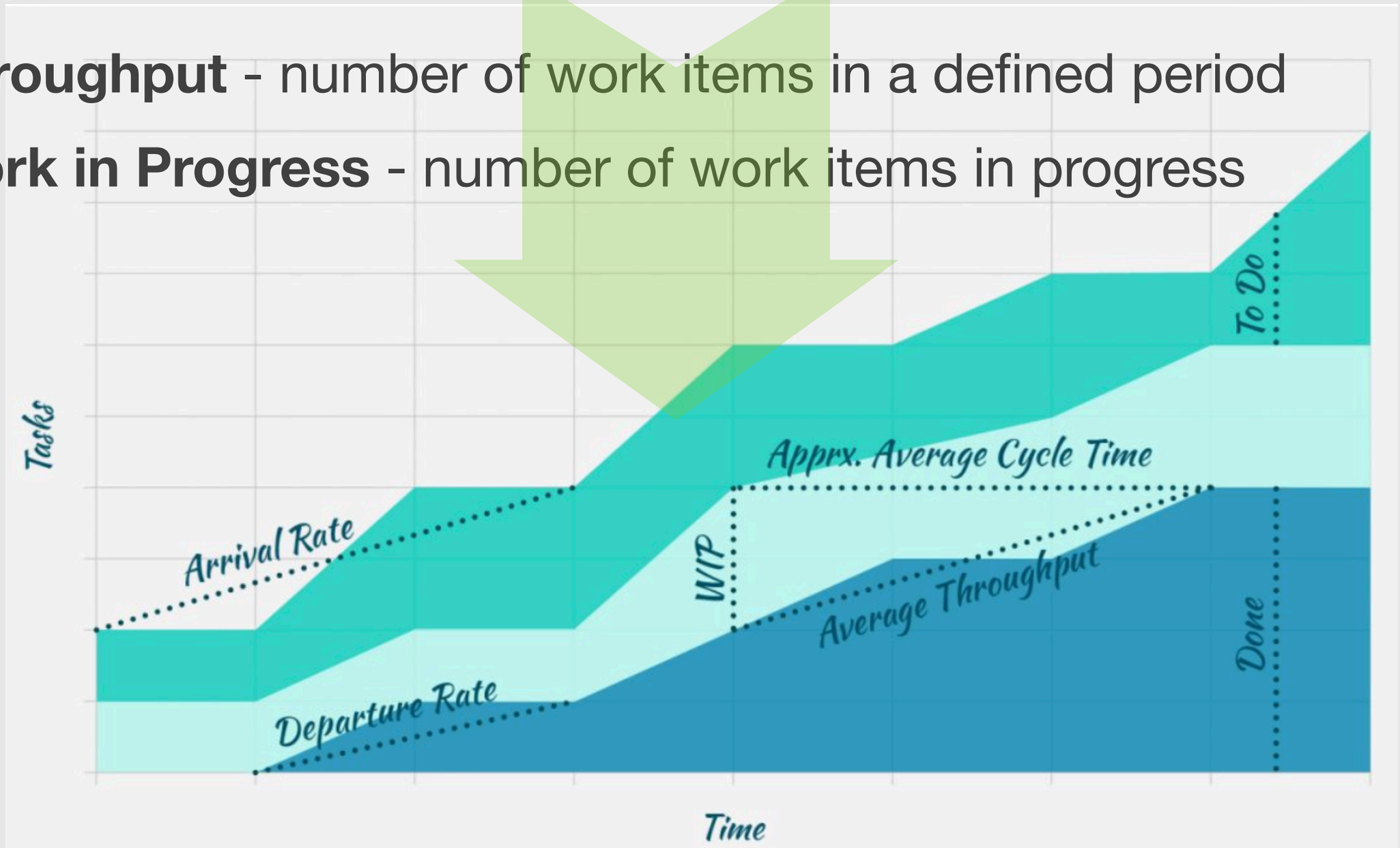


Kanban Board - Swim Lanes



Manage Work Flow - Metrics

- **Cycle Time** - average **<u>actual</u>** time for work item from start to finish
- **Throughput** - number of work items in a defined period
- **Work in Progress** - number of work items in progress



Software as a Product

SaaP

SaaP

- Application / Tool to solve a problem
- Usually one off purchase via a license (high price)
- Major updates = another purchase
- Traditional approach

SaaS

- Web App / Tool to solve a problem
- Subscription model (lower price) - pay as you go
- Dominate approach - 86% companies running SaaS

*https://www.singlegrain.com/saas/examples-of-saas-companies/

SaaS vs SaaP (Traditional Software)

- SaaS offers free trial
- Recurring payment
- No install needed
- Automatic updates
- Multiple users
- Manages security / data backups
- Distributed via cloud
  - Laptop, tablet, mobile